

WEEK 4 – EXPLORATORY DATA ANALYSIS

OBJECTIVE

The purpose of this analysis is to study the Shopify stock dataset using Python libraries such as Pandas, Matplotlib and Seaborn. The analysis focuses on stock prices, trading volume, moving averages, daily returns and stock volatility.

ANALYSIS

Analysis	Description
Stock Price Analysis	The Open, High, Low and Close values were examined to study changes in Shopify stock prices.
Volume Analysis	Trading volume was analyzed to observe the level of market activity on different dates.
Moving Average	20-day and 50-day moving averages were used to study the general direction of the closing price.
Price Change Analysis	A scatter plot was used to examine the relationship between trading volume and daily price change.
Daily Return Analysis	Mean, variance and standard deviation were used to understand daily stock returns.
Return Distribution	Histogram and KDE visualizations were used to observe the pattern of daily returns.
Volatility Analysis	The 90th percentile of absolute daily returns was used to detect comparatively high-volatility days.

KEY FINDINGS

- Shopify's stock prices fluctuate across different time periods.
- Trading volume shows variations in market activity.
- Moving averages help to identify the general trend in closing prices.
- Daily returns provide information about day-to-day price movement.
- Histogram and KDE plots show the overall pattern of stock returns.
- The volatility analysis highlights observations with relatively large daily movements.

MAJOR INSIGHTS

The EDA shows that Shopify stock performance can be understood by combining price, volume, return and moving-average analysis. The return and volatility analysis also helps identify periods where the stock experienced larger price changes.

RECOMMENDATIONS

- Monitor trading volume along with daily returns.
- Use moving averages to study changes in price trends.
- Examine high-volatility periods separately.
- Further analyze correlations between the available stock variables.

CONCLUSION

The Week 4 EDA provides an overview of Shopify stock behavior through statistical analysis and visualizations. The study of stock prices, volume, moving averages, daily returns and volatility helps in understanding important patterns in the dataset.